

Expt No:

02

WATER LEVEL CONTROL SYSTEM

Date:

/ /2025

1. AIM:

To design a ladder logic program to maintain the water level in a tank.

2. APPARATUS REQUIRED:

- o VPLCT - 01 [PLC Trainer kit].
- o Water level indicator [Float switches - 2].
- o Personal Computer [PC].
- o VersaPro Software Installed.
- o RS - 232 cable.
- o Patch Chords.

3. PROCEDURE:

- o Load the Versapro Software to the PC.
- o Open the Versapro Software.
- o Switch ON the PLC Trainer and Water level System.
- o Connect PLC and Water level System kit.
- o Open the New folder and draw the ladder logic program.
- o Connect the PLC to PC.
- o Select the correct hardware Configuration [Sl. No].
- o Store the Program to PLC.
- o Run the Program.
- o Verify the performance of the Water level System.

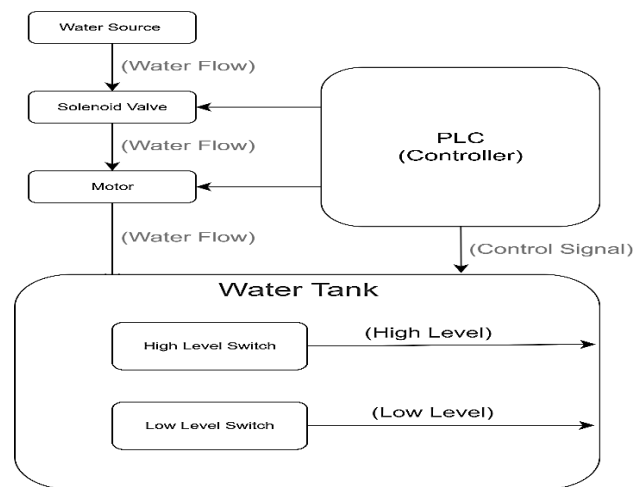
4. PROBLEM STATEMENT:

Water Level controller is used to Fill/drain the tank, which is depends upon high/low level in tank water. If the tank is in low level the solenoid valve will open & motor starts - ON Position if the tank is in high level, the solenoid valve will be close & motor stop - OFF position.

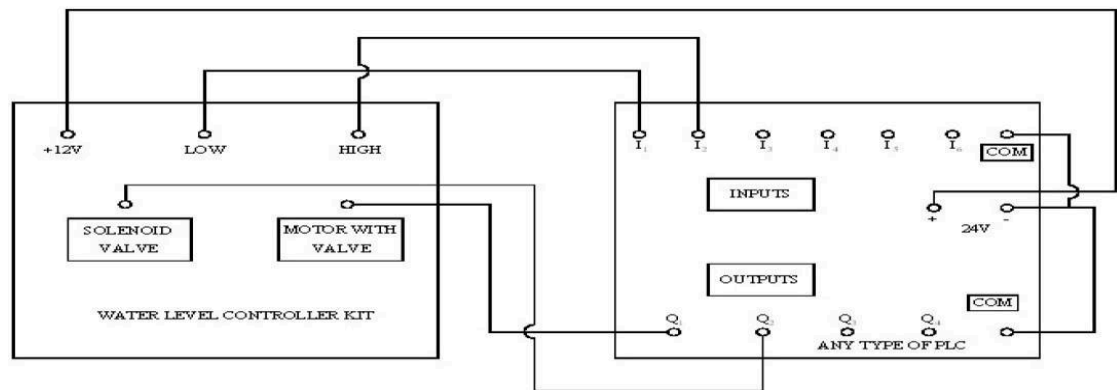
5. BLOCK DIAGRAM:



6. PROCESS DIAGRAM:



7. WIRING DIAGRAM:



8. LADDER DIAGRAM:



9. RESULT:

Thus, the ladder logic program to maintain the water level in a tank has been designed and tested using the water level kit.